

## REVIEW ARTICLE OPEN



# The effect of turmeric/curcumin supplementation on anthropometric indices in subjects with prediabetes and type 2 diabetes mellitus: A GRADE-assessed systematic review and dose-response meta-analysis of randomized controlled trials

Mohammadreza Moradi Baniasadi<sup>1,2,3</sup>, Pishva Arzhang<sup>4</sup>, Azin Setayesh<sup>2,3</sup>, Maedeh Moradi<sup>5,6</sup>, Ensieh Nasli-Esfahani<sup>1</sup> and Leila Azadbakht<sup>1,2,7</sup>

© The Author(s) 2025

The effective management of diabetes, a prevalent metabolic condition globally, relies on paying close attention to anthropometric measurements, while recent years have seen a growing interest in researching the potential anti-obesity properties of turmeric/curcumin. In this systematic review and dose-response meta-analysis of randomized controlled trials, the impact of turmeric/curcumin supplementation on anthropometric indices in individuals with prediabetes and type 2 diabetes mellitus (T2DM) was assessed. PubMed, Cochrane Library, Web of Science, Scopus, and Google Scholar were searched to identify relevant records published from 1 January 1990 to 1 June 2024. Random-effects meta-analysis was performed to evaluate the weighted mean difference (WMD) and 95% confidence interval (CI), with a  $p$ -value  $\leq 0.05$  indicating statistical significance. Grading of Recommendations Assessment, Development and Evaluation (GRADE) was used to assess the certainty of evidence. Twenty randomized controlled trials (RCTs) were included in the meta-analysis. Pooled analysis displayed that supplementation with turmeric/curcumin significantly decreased body weight (WMD:  $-1.9$  kg; 95% CI:  $-2.9$  to  $-0.9$ ;  $P \leq 0.001$ ; GRADE = low), waist circumference (WMD:  $-1.9$  cm; 95% CI:  $-3.5$  to  $-0.2$ ;  $P = 0.024$ ; GRADE = low), fat mass% (WMD:  $-2.9\%$ ; 95% CI:  $-5.6$  to  $-0.1$ ,  $P = 0.041$ ; GRADE = very low), and hip circumference (WMD:  $-1.0$  cm; 95% CI:  $-1.2$  to  $-0.8$ ;  $P \leq 0.001$ ; GRADE = moderate) but no effects on body mass index and waist-to-hip ratio in people with T2DM. In individuals with prediabetes, body weight (WMD:  $-2.5$  kg; 95% CI:  $-4.8$  to  $-0.2$ ;  $P = 0.037$ ; GRADE = moderate) and waist circumference (WMD:  $-2.9$  cm; 95% CI:  $-5.3$  to  $-0.6$ ;  $P = 0.015$ ; GRADE = moderate) were significantly lower in the turmeric/curcumin supplement-treated group than their untreated counterparts. The study found that turmeric/curcumin has a beneficial effect on some obesity indicators, which could contribute to weight management in individuals with prediabetes and T2DM. Systematic Review Registration: This study was registered at PROSPERO as CRD42023440612.

*Nutrition and Diabetes* (2025)15:34; <https://doi.org/10.1038/s41387-025-00386-7>

## INTRODUCTION

Type 2 diabetes mellitus (T2DM) represents a significant metabolic multifactorial disorder characterized by disruptions in pancreatic beta-cell function, peripheral tissue insulin resistance, and heightened pancreatic alpha-cell activity [1]. The risk of T2DM is influenced by abdominal obesity, irrespective of an individual's body mass index (BMI) [2]. In 2021, an estimated 536.6 million adults, constituting 10.5% of the global population, were afflicted with diabetes, with projections indicating an increase to 783.2 million individuals, or 12.2% of the population, by the year 2030 [3]. Diabetes imposes substantial societal burdens, encompassing

both direct and indirect consequences, such as diminished productivity and work-related losses [4]. According to the International Diabetes Federation, it is anticipated that the healthcare cost associated with diabetes will escalate to a staggering sum of United States Dollars (USD) 1.03 trillion by 2030 and USD 1.05 trillion by 2045 [5]. Diabetes can lead to damage in various organs, including the eyes, nerves, and kidneys, as well as fostering underlying cardiovascular ailments, stroke, and peripheral vascular disease [6]. Given the multifaceted nature of diabetes, with a multitude of risk factors at play, it is acknowledged that no single medication can comprehensively address all

<sup>1</sup>Diabetes Research Center, Endocrinology and Metabolism Clinical Sciences Institute, Tehran University of Medical Sciences, Tehran, Iran. <sup>2</sup>Department of Community Nutrition, School of Nutritional Sciences and Dietetics, Tehran University of Medical Sciences, Tehran, Iran. <sup>3</sup>Students' Scientific Research Center, Tehran University of Medical Sciences, Tehran, Iran. <sup>4</sup>Qods Hospital, Kermanshah University of Medical Sciences, Kermanshah, Iran. <sup>5</sup>Adelaide Medical School, The University of Adelaide, Adelaide, SA, Australia. <sup>6</sup>Lifelong Health Theme, South Australian Health and Medical Research Institute, Adelaide, SA, Australia. <sup>7</sup>Department of Community Nutrition, School of Nutrition and Food Science, Isfahan University of Medical Sciences, Isfahan, Iran. ✉email: [azadbakhtleila@gmail.com](mailto:azadbakhtleila@gmail.com)

Received: 22 August 2024 Revised: 18 May 2025 Accepted: 3 July 2025

Published online: 14 August 2025

its facets. Owing to the drawbacks associated with conventional drug therapies, such as side effects, high costs, and issues related to dosage tolerance, researchers are actively exploring novel, safe, and cost-effective alternatives for the management of diabetes [7].

Today herbal supplements for medicinal purposes are widely used, with around 80% of individuals incorporating them into their treatment plans [8]. Turmeric, a yellow pigment, derived from the *Curcuma longa* L. plant, belongs to the Zingiberaceae family and thrives in tropical climates [9]. It is commonly employed as a culinary additive in Southeast Asia to enhance the color and flavor of dishes. The active ingredient in turmeric is curcumin, also known as diferuloylmethane [10]. Turmeric ranks among the most frequently employed dietary supplements on a global scale [11]. Nevertheless, its therapeutic efficacy faces a notable challenge stemming from its low bioavailability, attributed to inadequate absorption, rapid metabolism, and swift elimination from the body. Consequently, recent endeavors have focused on enhancing its oral bioavailability through the development of novel drug delivery methodologies, including micelles, phospholipid complexes, nanoparticles, or the incorporation of adjunctive compounds [8]. Clinical evidence from both Phase I and II trials supports the safety of an 8 g/d curcumin dosage [12]. The bioavailability of turmeric can be significantly enhanced by the inclusion of piperine [13]. Research has indicated that the co-administration of just 20 milligrams of piperine with turmeric can result in a 20-fold increase in its absorption into the serum [13]. Increased bioavailability may potentially increase the risk of liver damage [14]. Research has demonstrated a remedial potential of curcumin in various conditions such as anti-obesity properties [8], anti-arthritis [15], ulcerative colitis [16], anti-inflammatory [17], antidepressant [18], liver function enhancer [19] lipid-lowering effect [10], anti-cytotoxicity [20], and other metabolic diseases [21]. Curcumin has beneficial effects on anthropometric parameters and glycemic indices in metabolic diseases [22]. Experimental studies have shown that curcumin decreases the expression of transcription factors that play a role in the production of fat in the liver [23]. Moreover, this substance can exert an anti-obesity influence by inhibiting mitogenesis and repressing the differentiation of preadipocytes [24].

Several meta-analyses have explored the impact of curcumin on anthropometric measures in both healthy and unhealthy populations [25–27]. Akbari et al. showed a significant reduction in BMI, weight, and waist circumference (WC) [25], although Ashtary-Larky did not observe similar outcomes [27]. The effect of curcumin on cardiovascular parameters in diabetes patients was evaluated by three meta-analyses [28–30]. Numerous randomized controlled trials (RCTs) have yielded promising results regarding the effect of curcumin/turmeric on body composition in individuals with diabetes [31] while other studies have reported negligible effects [32, 33]. Given the existing controversy surrounding the definitive impact of turmeric/curcumin on anthropometric indices in prediabetes and T2DM, we conducted a meta-analysis to arrive at a conclusive determination. To our knowledge, this study represents the comprehensive systematic review and dose-response meta-analysis aimed at assessing this particular influence.

## METHODS

We planned, implemented, and reported a systematic review and meta-analysis in compliance with Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [34]. A current systematic review and dose-response meta-analysis registered in the PROSPERO database (registration number: CRD42023440612).

## Search strategy

The current study employed a comprehensive search strategy utilizing various online databases, including PubMed, Cochrane

Library, Web of Science, Scopus, and Google Scholar, to identify relevant records published from 1 January 1990 to 1 June 2024. We chose the described databases to provide a comprehensive search. The exploration methodology implemented for each database is detailed in Supplementary Table 1. Additionally, a comprehensive manual search of reference lists of reviews about the topic was conducted to identify any additional relevant articles. In cases where necessary, the authors were contacted to obtain relevant data. It is important to note that studies that did not elicit a response from their respective authors were excluded from our analysis. There were no limitations on language.

## Study selection

The references were assessed for eligibility using a reference manager (EndNote X20; Thomson Reuters). Duplicates were removed and two reviewers (MMB and AS) separately screened the titles and abstracts. They then screened the full-text articles against inclusion and exclusion criteria. When there was disagreement, a third author (PA) was consulted to help reach a consensus through discussion.

Inclusion criteria: (1) studies with randomized controlled trials design (parallel or cross-over); (2) conducted on adults (>18 years old) with prediabetes and T2DM; (3) evaluated effect of turmeric and/or curcumin supplementation in comparison to placebo group on at least one of outcomes including body weight (BW), BMI, percentage of fat mass (FM %), WC, waist to hip ratio (WHR), and hip circumference (HC). We also considered studies that reported the changes in outcome variables.

Exclusion criteria: (1) studies with non-clinical trials and semi-experimental design; (2) performed on lactating or pregnant women, and animals; (3) insufficient data on our outcomes of interest; (4) articles that assessed the effect of curcumin/turmeric combined with other interventions.

## Data extraction

Two investigators (MMB and AS) independently extracted detailed information from each included study using a standardized template. A lead researcher (PA) was also present to adjudicate and resolve any discrepancies that might arise. In cases where published papers did not contain sufficient data, efforts were made to contact the corresponding author to procure the necessary information. We collected the following data points from each study: the name of the first author, publication year, study location, study design, demographic characteristics of participants (gender, age, BMI, health status), number of participants in intervention and placebo groups, study duration, type and dosage of intervention, type of placebo. The mean and standard deviation (SD) of parameters assessed at pre-, post-intervention besides the difference in outcomes between the beginning and end-trial were evaluated. Data from publications with the longest and largest intervention duration were extracted when multiple papers were published on the same study participants. Data was extracted from studies with interest outcomes reported in multiple intervals. Only baseline and intervention end data were considered.

## Quality assessment

Two investigators (MMB and AS) independently assessed the Risk of bias using the Cochrane tool [35]. The third author resolves any disagreements. The checklist for quality assessment comprises 7 criteria: (1) Random sequence generation, (2) Allocation concealment, (3) Blinding of participants and personnel, (4) Blinding of outcome assessment, (5) Incomplete outcome data, (6) Selective outcome reporting, (7) Other sources of bias. According to Cochrane Handbook guidelines, research studies were categorized into low, high, or unclear risk of bias for each criterion. Overall quality is calculated based on the below definition: (1) Good; if all items had “low risk”, (2) Fair;  $\leq 2$  items

rated as “unclear risk”, and (3) Poor; if  $\geq 1$  item had “high risk” or above three “unclear risk”.

### Statistical analysis

The study used mean change (SD) in body weight, BMI, FM%, HC, and WHR to calculate the mean difference (95% confidence interval [CI]) between intervention and placebo groups. When the SD and mean differences were not provided by the authors of the study, we counted them using the following formula: mean change = final values – baseline values; SD = square root  $[(SD \text{ baseline}^2 + SD \text{ final}^2) - (2 \times \text{correlation coefficient} \times SD \text{ baseline} \times SD \text{ final})]$  [36]. Based on eligible studies that have comprehensive data, we calculated the correlation coefficient (R). When studies reported standard errors (SEs), we calculated SD according to this formula:  $SD = SE \times \sqrt{n}$ , that ‘n’ was the number of participants in intervention and control groups. If the outcome is expressed in 95% CI, median, and interquartile range, we converted this to SD [37]. Statistical analyses were carried by STATA® version 17.0 (Stata Corp, College Station, Lakeway, TX, USA). In the presence of any heterogeneity, a random-effects model was used to calculate the pooled weighted mean difference (WMD) [38]. Between-study heterogeneity was assessed by the Cochran Q test and the I-square index. I-square over 50% with  $p$ -value  $< 0.05$  considered as significant heterogeneity [39]. Subgroup analyses were conducted in T2DM patients to detect possible sources of heterogeneity. Possible causes of heterogeneity were investigated using the following classifications: location (Iran vs. Other countries), duration of supplementation ( $\geq 12$  vs.  $< 12$  weeks), mean age ( $\geq 50$  vs.  $< 50$  years), baseline BMI ( $\geq 30$  vs.  $< 30 \text{ kg/m}^2$ ), dose ( $\geq 1000$  vs.  $< 1000 \text{ mg/day}$ ), and type of intervention (turmeric, unformulated curcumin, high absorption curcumin).

The analysis of publication bias for variables was conducted using the funnel plot test, Egger’s test, and Begg’s test [40, 41]. Fractional polynomial modeling was utilized to identify the non-linear impact of turmeric/curcumin dosage and duration on each outcome in the T2DM group. We conducted a sensitivity analysis to determine the influence of excluding a single study on the overall effect size [42].  $P < 0.05$  is considered statistically significant.

### Grading the evidence

The certainty of evidence is evaluated by Grading of Recommendations Assessment, Development, and Evaluation (GRADE) approach [43]. In brief, evidence obtained from randomized trials that were initially considered as having high certainty could be downgraded by a serious risk of bias, inconsistency, indirectness, imprecision, and evidence of publication bias. The evidence could also be upgraded due to the large effect size and the presence of a dose-response gradient. Two authors (MMB and PA) independently performed GRADE assessment. GRADE rates the certainty of evidence as high, moderate, low, and very low.

## RESULTS

### Literature search

Figure 1 illustrates the details of the selection procedure. Based on a primary search 9313 studies were found. After removing duplicate citations, 6342 remained for title/abstract evaluation. Out of 6342, 65 Studies entered to full-text check. Eventually, 20 eligible articles were taken in our Systematic Review and Dose-response Meta-analysis.

### Study characteristics

The features of 20 eligible studies are described in Table 1. In general, 20 RCTs, with 1387 participants (698 cases and 689 controls) involved. The range dose of turmeric/curcumin was from 80 mg/day

[44–48] to 2100 mg/day [49–51]. The duration of RCTs varied from 8 weeks [45, 49–53] to 36 weeks [31]. The type of intervention is divided into unformulated curcumin [31, 33, 52, 54–58], high-absorption curcumin [32, 44–48, 53, 59, 60], and turmeric [49–51]. The subjects enrolled in this systematic review and meta-analysis (SRMA) had either Prediabetes [31–33, 53, 58] or T2DM. We gathered mean and corresponding SD of outcomes of interest in two studies via sending email to corresponding authors [44, 52]. The language of the included studies was English. The percentage of FM in three study measured using bioelectrical impedance analysis [50, 54, 57].

### Quality assessment

The detail of quality evaluation is exhibited in Fig. 2. Among 20 studies, five RCTs were considered good quality [31, 45, 46, 53, 58]. Eleven trials ranked as fair quality [32, 33, 44, 47–49, 54–57, 60] and four studies had poor quality [50–52, 59].

### Effect of turmeric/curcumin on BW in T2DM patients

Totally 14 RCTs were performed on BW in T2DM patients with 479 and 470 participants in intervention and placebo groups, respectively [44–50, 52, 54–57, 59, 60]. Pooled results of these RCTs showed a significant decrease in BW (WMD:  $-1.9 \text{ kg}$ ; 95% CI:  $-2.9$  to  $-0.9$ ;  $P \leq 0.001$ ; GRADE = low) with significant heterogeneity across studies ( $I^2 = 89.6\%$ ;  $P \leq 0.001$ ) (Fig. 3). Subgroup analyses indicated supplementation with turmeric (WMD:  $-5.1 \text{ kg}$ ; 95% CI:  $-5.9$  to  $-4.3$ ;  $P \leq 0.001$ ) in participants with BMI  $< 30 \text{ kg/m}^2$  (WMD:  $-2.2 \text{ kg}$ ; 95%CI:  $-3.4$  to  $-0.9$ ;  $P = 0.001$ ) significantly lowered BW (Table 2).

### Effect of turmeric/curcumin on BMI in T2DM patients

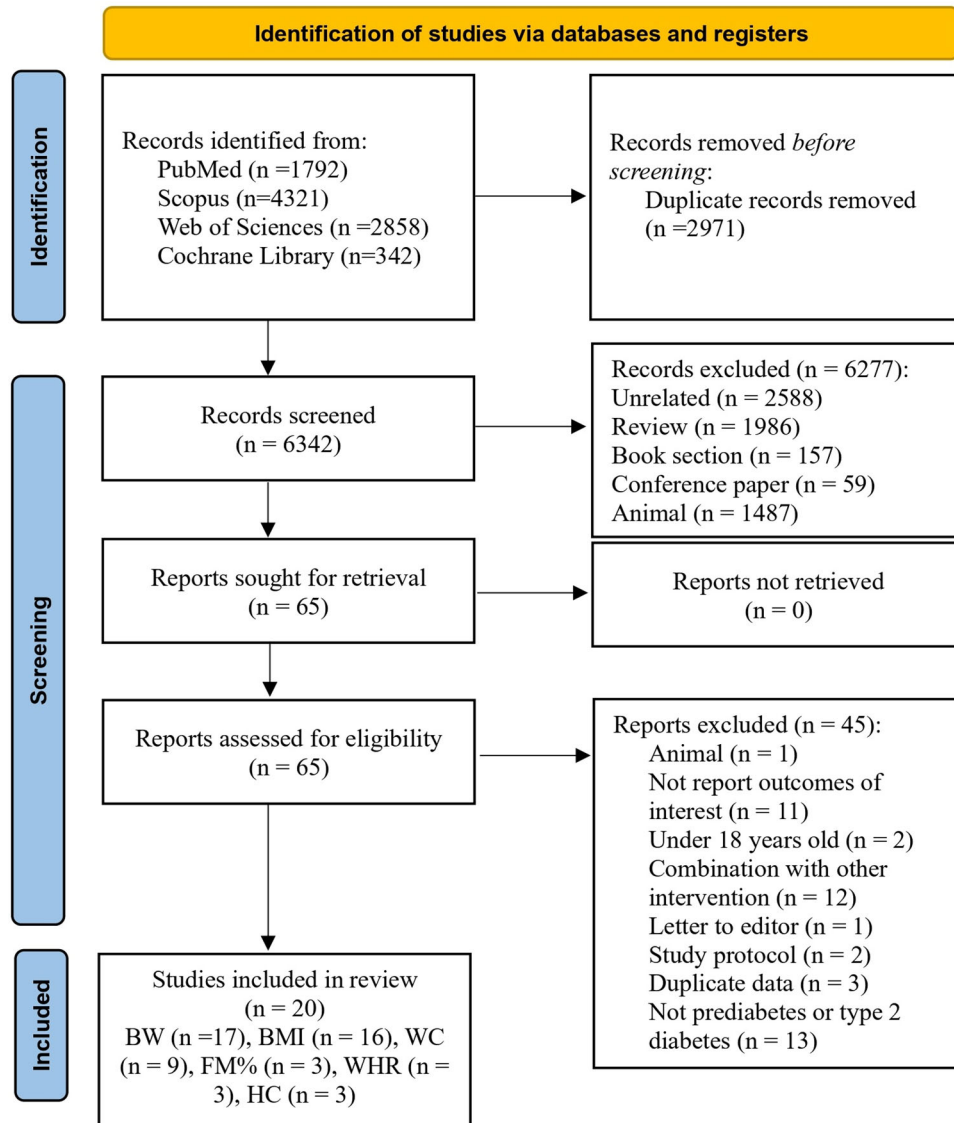
Thirteen trials with 952 participants (456 cases and 446 controls) evaluated BMI in T2DM patients [44–47, 49, 50, 52, 54–57, 59, 60]. Combined effect size revealed that turmeric/curcumin did not significantly decrease BMI (WMD:  $-0.5 \text{ kg/m}^2$ ; 95% CI:  $-1.3$  to  $0.2$ ;  $P = 0.173$ ; GRADE: very low). However, high heterogeneity among RCTs ( $I^2 = 99.2\%$ ;  $P \leq 0.001$ ) was seen (Fig. 4). Results of subgroup analyses showed that supplementation with turmeric (WMD:  $-1.7 \text{ kg/m}^2$ ; 95% CI:  $-3.2$  to  $-0.1$ ;  $P = 0.039$ ) and high-absorption curcumin (WMD:  $-0.5 \text{ kg/m}^2$ ; 95% CI:  $-0.8$ ,  $-0.2$ ;  $P = 0.004$ ) for  $\geq 12$  weeks (WMD:  $-0.7 \text{ kg/m}^2$ ; 95% CI:  $-1.0$  to  $-0.3$ ;  $P \leq 0.001$ ) significantly reduced BMI in sample size  $\geq 60$  (WMD:  $-0.7 \text{ kg/m}^2$ ; 95% CI:  $-1.1$  to  $-0.4$ ;  $P \leq 0.001$ ) (Table 2). However,  $P$ -value for difference between these subgroups is not significant.

### Effect of turmeric/curcumin on WC in T2DM patients

Findings from five RCTs [45, 51, 54, 55, 60] in 419 T2DM patients (212 cases and 207 controls) showed that turmeric/curcumin supplementation versus placebo group favorable reduced WC (WMD:  $-1.9 \text{ cm}$ ; 95% CI:  $-3.5$  to  $-0.2$ ;  $P = 0.024$ ; GRADE: low) while between-study heterogeneity was high ( $I^2 = 96.1\%$ ;  $P \leq 0.001$ ) (Fig. 5). When operating subgroup analysis, we observed a significant reduction of WC in trials with good (WMD:  $-1.4 \text{ cm}$ ; 95% CI:  $-2.3$  to  $-0.5$ ;  $P = 0.002$ ) and poor quality (WMD:  $-3.6 \text{ cm}$ ; 95% CI:  $-4.1$  to  $-3.1$ ;  $P \leq 0.001$ ). Additionally, supplementation with turmeric (WMD:  $-3.6 \text{ cm}$ ; 95% CI:  $-4.1$  to  $-3.1$ ;  $P \leq 0.001$ ) and high-absorption curcumin (WMD:  $-1.3 \text{ cm}$ ; 95% CI:  $-2.2$  to  $-0.4$ ;  $P = 0.003$ ) significantly lowered WC in T2DM patients (Table 2).

### Effect of turmeric/curcumin on FM % in T2DM patients

The impact of turmeric/curcumin on FM% was detected by three RCTs [50, 54, 57] which included total of 269 individuals (134 in intervention group and 135 in the placebo group). The meta-analysis of these RCTs indicated that FM% was significantly decreased following turmeric/curcumin consumption (WMD:  $-2.9\%$ ; 95% CI:  $-5.6$  to  $-0.1$ ,  $P = 0.041$ ; GRADE = very low) and significant heterogeneity was seen across studies ( $I^2 = 84.6\%$ ;  $P = 0.002$ ) (Fig. 6).



**Fig. 1** PRISMA 2020 Flow diagram of the selection study process. BW body weight, BMI body mass index, WC waist circumference; FM% fat mass%; WHR waist-to-hip ratio; HC hip circumference.

#### Effect of turmeric/curcumin on HC in T2DM patients

The findings of two RCTs (comprising 54 cases and 51 controls) [55, 60] showed significant alterations in the HC in T2DM patients (WMD:  $-1.0$  cm; 95% CI:  $-1.2$  to  $-0.8$ ;  $P \leq 0.001$ ; GRADE = moderate) with non-significant heterogeneity among studies ( $I^2 = 0.0\%$ ;  $P = 0.560$ ) (Fig. 7).

#### Effect of turmeric/curcumin on WHR in T2DM patients

A combination of 3 effect sizes with 117 individuals [50, 57, 60] revealed turmeric/curcumin intake did not change WHR in T2DM patients (WMD:  $-0.1$ ; 95% CI:  $-0.4$  to  $0.1$ ;  $P = 0.398$ ; GRADE = very low) although there was significant heterogeneity between trials ( $I^2 = 99.7\%$ ;  $P \leq 0.001$ ) (Fig. 8).

#### Effect of turmeric/curcumin on BW in prediabetes

According to inclusion criteria, combined results of 3 RCTs with 304 participants [31, 33, 58] reported a favorable effect of turmeric/curcumin supplementation on BW in prediabetes (WMD:  $-2.5$  kg; 95% CI:  $-4.8$  to  $-0.2$ ;  $P = 0.037$ ; GRADE = moderate) and significant heterogeneity between studies was detected ( $I^2 = 77.6\%$ ;  $P = 0.012$ ) (Supplementary Fig. 1).

#### Effect of turmeric/curcumin on BMI in prediabetes

Mixed results of three publications [32, 53, 58] based on a random-effects model showed BMI in prediabetes did not change after consumption of turmeric/curcumin (WMD:  $-0.5$  kg/m<sup>2</sup>; 95% CI:  $-1.2$  to  $0.1$ ;  $P = 0.113$ ; GRADE = low). Moreover, we did not observe significant heterogeneity between RCTs ( $I^2 = 50.5\%$ ;  $P = 0.133$ ) (Supplementary Fig. 2).

#### Effect of turmeric/curcumin on WC in prediabetes

The results of the meta-analysis indicated that turmeric/curcumin has a lowering effect on WC across the four trials [31, 33, 53, 58] in prediabetes (WMD:  $-2.9$  cm; 95% CI:  $-5.3$  to  $-0.6$ ;  $P = 0.015$ ;  $I^2 = 81.0\%$ ;  $P_{\text{heterogeneity}} = 0.001$ ; GRADE = moderate) (Supplementary Fig. 3).

#### Sensitivity analysis

We stepwise leave-out each trial to evaluate the influence of each study on the total result. Results of sensitivity analysis revealed that overall effect sizes of BW, BMI, and WC were not influenced by individual studies in T2DM patients.

**Table 1.** Characteristics of included studies.

| Author              | Year | Location  | Study design | Health status                                     | Gender (M/F) | Sample size |     | Age (mean ± SD) |               | BMI (mean ± SD) |              | Intervention                                                    |                                                    |
|---------------------|------|-----------|--------------|---------------------------------------------------|--------------|-------------|-----|-----------------|---------------|-----------------|--------------|-----------------------------------------------------------------|----------------------------------------------------|
|                     |      |           |              |                                                   |              | IG          | CG  | IG              | CG            | IG              | CG           | Treatment group                                                 | Control group                                      |
| Chuengsamarn [31]   | 2012 | Thailand  | RCT, DB, Pa  | Prediabetes                                       | 82/152       | 118         | 116 | 56.95 ± 11.94   | 57.93 ± 12.7  | 26.66 ± 5.21    | 26.62 ± 5.49 | Curcumin (1500 mg/day)                                          | Placebo                                            |
| Chuengsamarn [54]   | 2014 | Thailand  | RCT, DB, Pa  | T2DM                                              | 97/116       | 107         | 106 | 59.16 ± 11.03   | 59.58 ± 10.7  | 27.09 ± 5.37    | 26.84 ± 4.32 | Curcumin (1500 mg/day)                                          | Starch                                             |
| Jimenez-Osorio [52] | 2016 | Mexico    | RCT, DB, Pa  | Diabetic proteinuric CKD                          | 36/15        | 28          | 23  | 55.0 ± 8.46     | 56.2 ± 7.19   | 29.7 ± 6.34     | 27.9 ± 5.27  | Curcumin (320 mg/day)                                           | Starch                                             |
| Rahimi [44]         | 2016 | Iran      | RCT, DB, Pa  | T2DM                                              | 31/39        | 35          | 35  | 56.34 ± 11.17   | 60.95 ± 10.77 | 26.92 ± 2.71    | 27.27 ± 3.59 | Nano-curcumin (80 mg/day)                                       | Placebo                                            |
| Panahi [59]         | 2018 | Iran      | RCT, DB, Pa  | T2DM                                              | 51/49        | 50          | 50  | 43 ± 8          | 41 ± 7        | 26 ± 2          | 27 ± 2       | Curcuminoids [Curcumin C3 Complex*] (500 mg/day+ 5 mg piperine) | Lactose + 5 mg piperine                            |
| Adab [49]           | 2019 | Iran      | RCT, DB, Pa  | Hyperlipidemic T2DM                               | 36/39        | 39          | 36  | 54.76 ± 6       | 55.66 ± 8.64  | 28.98 ± 3.68    | 28.82 ± 4.96 | Turmeric powder (2100 mg/day)                                   | Corn starch flour                                  |
| Asadi [45]          | 2019 | Iran      | RCT, DB, Pa  | T2DM                                              | 10/70        | 40          | 40  | 53.3 ± 6.5      | 54.6 ± 6.2    | 31.1 ± 4.2      | 30.8 ± 3.8   | Nano-curcumin (80 mg/day)                                       | Polysorbate                                        |
| Funamoto [32]       | 2019 | Japan     | RCT, DB, Pa  | IGT                                               | 23/10        | 15          | 18  | 70 ± 6          | 69 ± 7        | 24.9 ± 4.6      | 25 ± 2.6     | Theracurmin (180 mg/day)                                        | Placebo                                            |
| Hodaei [55]         | 2019 | Iran      | RCT, DB, Pa  | T2DM                                              | 22/22        | 21          | 23  | 58 ± 8          | 60 ± 7        | 29.2 ± 3.76     | 28.2 ± 2.5   | Curcumin (1500 mg/day)                                          | Cooked rice flour                                  |
| Cicero [53]         | 2020 | Italy     | RCT, DB, Pa  | Overweight subjects with suboptimal values of FPG | 37/43        | 40          | 40  | 54 ± 3          | 53 ± 5        | 27.1 ± 1.8      | 26.9 ± 1.9   | Phytosomal curcumin (1600 mg/day)                               | Placebo                                            |
| Ebrahimkhani [57]   | 2020 | Iran      | RCT, DB, Pa  | T2DM + hypovitaminosis D                          | 22/13        | 16          | 19  | 56.19           | 52.16         | 31.8 ± 4.94     | 30.3 ± 4.7   | Curcuminoids (500 mg/day)                                       | Maltodextrin                                       |
| Shafabakhsh_a [56]  | 2020 | Iran      | RCT, DB, Pa  | T2DM + CHD                                        | NR           | 25          | 24  | 64.9 ± 7.8      | 66.5 ± 7.7    | 30.3 ± 5.9      | 29.8 ± 2.9   | Curcumin (1000 mg/day)                                          | Starch                                             |
| Shafabakhsh_b [46]  | 2020 | Iran      | RCT, DB, Pa  | Diabetes on hemodialysis                          | 32/21        | 26          | 27  | 58.3 ± 9.4      | 56.2 ± 9.8    | 27.9 ± 4.9      | 27.1 ± 4.2   | Nano-curcumin (80 mg/day)                                       | Placebo                                            |
| Thota [33]          | 2020 | Australia | RCT, DB, Pa  | IFG/IGT                                           | 12/17        | 14          | 15  | 54.5 ± 10.85    | 50.4 ± 10.06  | 30.2 ± 4.11     | 32.3 ± 6.58  | Curcumin [Meriva*] (1000 mg/day)                                | Placebo                                            |
| Darmian [50]        | 2021 | Iran      | RCT, SB, Pa  | Hyperlipidemic T2DM                               | 0/21         | 11          | 10  | 44.33 ± 1.23    | 44.22 ± 3.07  | 29.3 ± 0.3      | 29.02 ± 1.04 | Turmeric powder (2100 mg/day)                                   | Corn starch flour                                  |
| Mokhtari [47]       | 2021 | Iran      | RCT, DB, Pa  | T2DM + DFU                                        | 39/11        | 25          | 25  | 57.4 ± 11.7     | 55.8 ± 9.4    | 27.5 ± 4.9      | 30.2 ± 5.3   | Nano-curcumin (80 mg/day)                                       | Placebo                                            |
| Sousa [60]          | 2021 | Brazil    | RCT, DB, Pa  | T2DM                                              | 14/47        | 33          | 28  | 63.2 ± 11.1     | 61.9 ± 11     | 29.66 ± 4.8     | 28.58 ± 5    | Long turmeric (500 mg/day+ 5 mg piperine)                       | Carboxymethyl cellulose                            |
| Darmian [51]        | 2022 | Iran      | RCT, SB, Pa  | Hyperlipidemic T2DM                               | 0/21         | 11          | 10  | 44.33 ± 1.23    | 44.22 ± 3.07  | 29.3 ± 0.3      | 29.02 ± 1.04 | Turmeric powder (2100 mg/day)                                   | Corn starch flour                                  |
| Karandish [58]      | 2022 | Iran      | RCT, DB, Pa  | IGT                                               | 13/28        | 21          | 20  | 36.95 ± 7.23    | 34.19 ± 7.03  | 30.46 ± 2.75    | 30.97 ± 2.33 | Curcumin (500 mg/day)                                           | Roasted rice powder                                |
| Asghari [48]        | 2024 | Iran      | RCT, DB, Pa  | T2DM                                              | 23/27        | 23          | 24  | 54.56 ± 8.30    | 57.48 ± 11.27 | 27.78 ± 2.15    | 27.82 ± 1.73 | Nano-curcumin and placebo of ω-3 Fatty Acids (80 mg/day)        | ω-3 fatty acids placebos and nano curcumin placebo |

BMI body mass index, M male, F female, IG intervention group, CG control group, RCT randomized controlled trial, DB double-blind, SB single-blind, Pa parallel, T2DM type 2 diabetes mellitus, CKD chronic kidney disease, IGT impaired glucose tolerance, FPG fasting plasma glucose, CHD coronary heart disease, IFG impaired fasting glucose, DFU diabetic foot ulcer.



|                       | Risk of bias |    |    |    |    |    |    |
|-----------------------|--------------|----|----|----|----|----|----|
|                       | D1           | D2 | D3 | D4 | D5 | D6 | D7 |
| Chuengsamarn (2012)   | +            | +  | +  | +  | +  | +  | +  |
| Chuengsamarn (2014)   | +            | +  | +  | -  | +  | +  | +  |
| Jimenez-Osorio (2016) | +            | -  | +  | -  | +  | +  | -  |
| Rahimi (2016)         | +            | +  | +  | +  | -  | +  | -  |
| Panahi (2018)         | +            | +  | +  | -  | -  | +  | -  |
| Adab (2019)           | +            | +  | +  | -  | +  | +  | +  |
| Asadi (2019)          | +            | +  | +  | +  | +  | +  | +  |
| Funamoto (2019)       | +            | +  | +  | +  | -  | -  | -  |
| Hodaei (2019)         | +            | +  | +  | -  | +  | +  | +  |
| Cicero (2020)         | +            | +  | +  | +  | +  | +  | +  |
| Ebrahimkhani (2020)   | +            | +  | +  | -  | +  | +  | -  |
| Shafabakhsh_a (2020)  | +            | -  | +  | +  | +  | +  | -  |
| Shafabakhsh_b (2020)  | +            | +  | +  | +  | +  | +  | +  |
| Thota (2020)          | +            | +  | +  | +  | -  | +  | -  |
| Darmian (2021)        | +            | +  | +  | X  | -  | -  | -  |
| Mokhtari (2021)       | +            | +  | +  | -  | +  | +  | -  |
| Sousa (2021)          | +            | +  | +  | +  | +  | +  | -  |
| Dramian (2022)        | +            | +  | +  | X  | -  | -  | -  |
| Karandish (2022)      | +            | +  | +  | +  | +  | +  | +  |
| Asghari (2024)        | +            | +  | +  | +  | +  | -  | -  |

Study

D1: Random sequence generation  
D2: Allocation concealment  
D3: Blinding of participants and personnel  
D4: Blinding of outcome assessment  
D5: Incomplete outcome data  
D6: Selective reporting  
D7: Other sources of bias

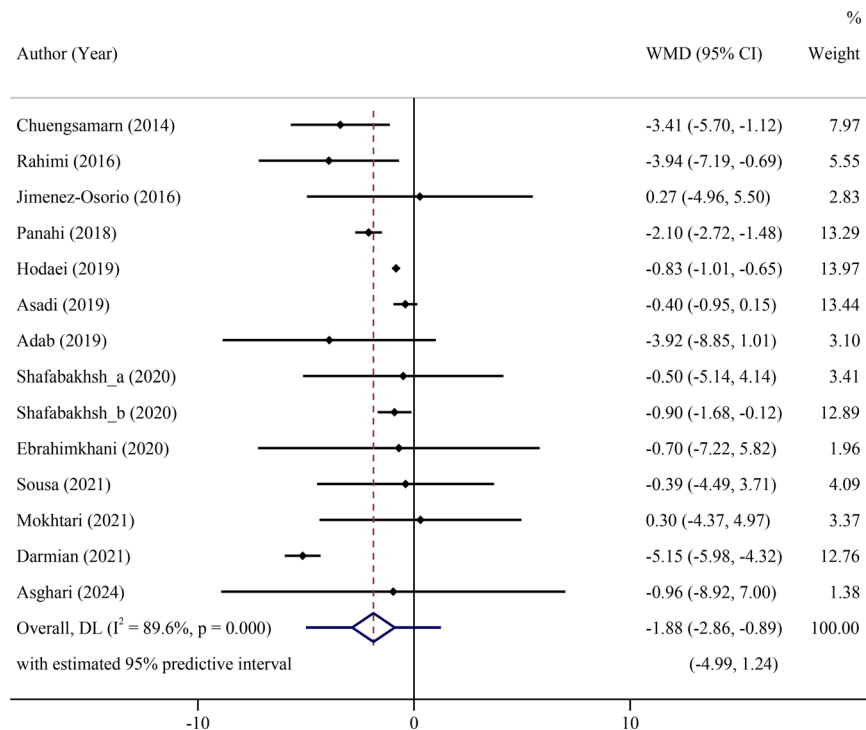
Judgement  
X High  
- Unclear  
+ Low

**Fig. 2 Results of risk of bias evaluation according to the Cochrane tool.** The figure summarizes the assessment of studies across different bias domains, with ratings classified as low, unclear, or high risk.

#### Publication bias

We use funnel plot, Begg's test, and Egger's test for detection of publication bias. There was no observable publication bias for BW

( $P = 0.26$ , Egger's test and  $P = 0.83$ , Begg's test), BMI ( $P = 0.14$ , Egger's test and  $P = 0.20$ , Begg's test), and WC ( $P = 0.41$ , Egger's test and  $P = 1.00$ , Begg's test) in T2DM patients (Supplementary Fig. 4).



**Fig. 3 Forest plot of effect of turmeric/curcumin on body weight in people with type 2 diabetes mellitus.** Horizontal lines represent 95% confidence intervals. Diamonds represent pooled estimates from random-effects analysis. Red dotted line represents null value. WMD: weighted mean difference; DL: DerSimonian & Laird method for random-effect meta-analysis.

### Dose-response analysis in T2DM

Based on dose-response analysis, a significant association between the dosage of turmeric/curcumin and WC ( $P_{\text{non-linearity}} = 0.04$ ) was seen. We failed to find a significant relation between turmeric/curcumin dose and BW ( $P_{\text{non-linearity}} = 0.06$ ) and BMI ( $P_{\text{non-linearity}} = 0.45$ ) (Fig. 9). Likewise, the dose-response analysis indicated a non-significant association between duration of turmeric/curcumin supplementation with BMI ( $P_{\text{non-linearity}} = 0.08$ ) and WC ( $P_{\text{non-linearity}} = 0.51$ ). However, the linkage between duration of turmeric/curcumin consumption and BW was significant ( $P_{\text{non-linearity}} = 0.01$ ) (Fig. 10). Turmeric/curcumin supplementation with a duration > 22 weeks ( $\text{WMD} = -2.5$  kg) revealed a significant reduction in BW. Moreover, a significant lowering effect of turmeric/curcumin supplementation on WC was observed at a dosage of > 1500 mg/d ( $\text{WMD} = -1.8$  cm).

### Adverse events

Three of the twenty included RCTs reported few side effects. In the first RCT, Supplementation with 80 mg/d nano-curcumin led to stomachache in two people with T2DM [45]. Chuengsamarn et al. observed itching, vertigo, and constipation with 1500 mg/d curcumin supplementation [31]. hot flash, constipation and nausea were reported by one RCT that administrated 1500 mg/d curcumin for 6 months [54].

### Grading the evidence

Details of the certainty assessment are depicted in Supplementary Table 2.

### DISCUSSION

In this systematic review and dose-response meta-analysis, we evaluated the impact of turmeric/curcumin on anthropometric indices in individuals with prediabetes and patients with T2DM by pooling the results of 20 RCTs. Our analysis revealed that turmeric/

curcumin ultimately led to significant improvements in BW, WC, FM%, and HC among T2DM patients. However, association between turmeric/curcumin intake and BMI, WHR was not significant in these patients. In prediabetes conditions, we realized administration of turmeric/curcumin significantly reduced BW and WC but did not affect BMI. In a non-linear dose-response fashion, a significant association between dosage of turmeric/curcumin with and WC and between duration of turmeric/curcumin consumption and BW was observed.

This systematic review and dose-response meta-analysis has elucidated that the intake of turmeric/curcumin exerts a positive influence on BW and WC among both prediabetes and T2DM patients. Our findings align with those of SRMA that demonstrated a reduction in BW and WC in adults following curcumin/turmeric supplementation [61]. Contrary to our results, a meta-analysis observed turmeric or curcumin did not affect BW and WC in patients with non-alcoholic fatty liver disease [62]. Moreover, two studies reported no lowering effect of curcumin on WC in adults and metabolic diseases [22, 26]. In a trial by Adab et al., turmeric did not change BW significantly in people with hyperlipidemia and T2DM [49]. In the mentioned study, a small sample size and short duration were assumed as a limitation of the RCT. Based on subgroup analysis, BW-lowering properties of turmeric/curcumin were observed in participants with  $\text{BMI} < 30 \text{ kg/m}^2$  and were administered with turmeric and high-absorption curcumin. Results of subgroup analysis imply turmeric/curcumin administration significantly mitigated WC in participants with < 50 years and in forms of turmeric and high absorption curcumin. Due to the importance of waist circumference in diabetes prognosis [63], there will be required RCTs with a large sample size.

In our systematic review and dose-response meta-analysis, it is evident that turmeric/curcumin leads to a significant reduction in FM% and HC among patients with T2DM. Dehzad et al. showed that curcumin/turmeric consumption effectively reduced FM% in adults, which affirms our result [61]. Additionally, Hodaei et al.

**Table 2.** Results of subgroup analysis for the effects of turmeric/curcumin supplementation on anthropometric indices in patients with T2DM.

|                             | Number of studies | WMD (95% CI)         | P-value <sup>a</sup> | I <sup>2</sup> (%) <sup>b</sup> | P for heterogeneity | P between Subgroups <sup>c</sup> |
|-----------------------------|-------------------|----------------------|----------------------|---------------------------------|---------------------|----------------------------------|
| <b>Body weight</b>          |                   |                      |                      |                                 |                     |                                  |
| Overall                     | 14                | −1.88 (−2.86, −0.89) | ≤0.001               | 89.6                            | ≤0.001              |                                  |
| <b>Age</b>                  |                   |                      |                      |                                 |                     |                                  |
| <50 years                   | 2                 | −3.61 (−6.60, −0.62) | 0.018                | 97.0                            | ≤0.001              | 0.069                            |
| ≥50 years                   | 12                | −0.82 (−1.16, −0.49) | ≤0.001               | 13.4                            | 0.314               |                                  |
| <b>Dose</b>                 |                   |                      |                      |                                 |                     |                                  |
| <1000 mg                    | 9                 | −1.18 (−2.04, −0.31) | 0.008                | 60.2                            | 0.010               | 0.265                            |
| ≥1000 mg                    | 5                 | −2.83 (−5.60, −0.06) | 0.045                | 96.2                            | ≤0.001              |                                  |
| <b>BMI</b>                  |                   |                      |                      |                                 |                     |                                  |
| <30 kg/m <sup>2</sup>       | 11                | −2.19 (−3.44, −0.94) | 0.001                | 91.6                            | ≤0.001              | 0.010                            |
| ≥30 kg/m <sup>2</sup>       | 3                 | −0.40 (−0.95, 0.14)  | 0.147                | 0.0                             | 0.995               |                                  |
| <b>Duration</b>             |                   |                      |                      |                                 |                     |                                  |
| <12 weeks                   | 5                 | −2.05 (−3.88, −0.23) | 0.027                | 96.2                            | ≤0.001              | 0.734                            |
| ≥12 weeks                   | 9                 | −1.71 (−2.49, −0.93) | ≤0.001               | 27.7                            | 0.198               |                                  |
| <b>Type Of Intervention</b> |                   |                      |                      |                                 |                     |                                  |
| Turmeric                    | 2                 | −5.12 (−5.93, −4.30) | ≤0.001               | 0.0                             | 0.630               | ≤0.001                           |
| Unformulated curcumin       | 5                 | −1.22 (−2.34, −0.10) | 0.032                | 20.6                            | 0.283               |                                  |
| High absorption curcumin    | 7                 | −1.23 (−2.17, −0.30) | 0.010                | 69.7                            | 0.003               |                                  |
| <b>Location</b>             |                   |                      |                      |                                 |                     |                                  |
| Iran                        | 11                | −1.88 (−2.95, −0.80) | 0.001                | 91.7                            | ≤0.001              | 0.938                            |
| Other Countries             | 3                 | −1.98 (−4.34, 0.39)  | 0.101                | 25.0                            | 0.264               |                                  |
| <b>Risk of bias</b>         |                   |                      |                      |                                 |                     |                                  |
| Good                        | 2                 | −0.57 (−1.03, −0.11) | 0.016                | 4.6                             | 0.306               | 0.074                            |
| Fair                        | 9                 | −1.53 (−2.57, −0.48) | 0.004                | 20.9                            | 0.257               |                                  |
| Poor                        | 3                 | −2.97 (−5.69, −0.25) | 0.032                | 94.3                            | ≤0.001              |                                  |
| <b>Body Mass Index</b>      |                   |                      |                      |                                 |                     |                                  |
| Overall                     | 13                | −0.52 (−1.27, 0.23)  | 0.173                | 99.2                            | ≤0.001              |                                  |
| <b>Age</b>                  |                   |                      |                      |                                 |                     |                                  |
| <50 years                   | 2                 | −1.58 (−3.31, 0.14)  | 0.072                | 99.4                            | ≤0.001              | 0.144                            |
| ≥50 years                   | 11                | −0.26 (−0.65, 0.12)  | 0.176                | 90.7                            | ≤0.001              |                                  |
| <b>Dose</b>                 |                   |                      |                      |                                 |                     |                                  |
| <1000 mg                    | 8                 | −0.21 (−0.69, 0.26)  | 0.381                | 86.9                            | ≤0.001              | 0.376                            |
| ≥1000 mg                    | 5                 | −0.85 (−2.25, 0.54)  | 0.232                | 99.7                            | ≤0.001              |                                  |
| <b>BMI</b>                  |                   |                      |                      |                                 |                     |                                  |
| <30 kg/m <sup>2</sup>       | 10                | −0.60 (−1.52, 0.31)  | 0.197                | 99.4                            | ≤0.001              | 0.363                            |
| ≥30 kg/m <sup>2</sup>       | 3                 | −0.17 (−0.39, 0.05)  | 0.140                | 0.0                             | 0.999               |                                  |
| <b>Duration</b>             |                   |                      |                      |                                 |                     |                                  |
| <12 weeks                   | 5                 | −0.40 (−1.72, 0.91)  | 0.548                | 99.7                            | ≤0.001              | 0.709                            |
| ≥12 weeks                   | 8                 | −0.66 (−1.00, −0.33) | ≤0.001               | 46.0                            | 0.073               |                                  |
| <b>Type Of Intervention</b> |                   |                      |                      |                                 |                     |                                  |
| Turmeric                    | 2                 | −1.66 (−3.24, −0.08) | 0.039                | 98.7                            | ≤0.001              | 0.154                            |
| Unformulated curcumin       | 5                 | 0.05 (−0.77, 0.88)   | 0.898                | 86.3                            | ≤0.001              |                                  |
| High absorption curcumin    | 6                 | −0.48 (−0.81, −0.15) | 0.004                | 70.9                            | 0.004               |                                  |
| <b>Location</b>             |                   |                      |                      |                                 |                     |                                  |
| Iran                        | 10                | −0.67 (−1.53, 0.20)  | 0.131                | 99.4                            | ≤0.001              | 0.554                            |
| Other Countries             | 3                 | −0.04 (−1.93, 1.85)  | 0.967                | 92.8                            | ≤0.001              |                                  |
| <b>Risk of bias</b>         |                   |                      |                      |                                 |                     |                                  |
| Good                        | 2                 | −0.23 (−0.40, −0.06) | 0.008                | 0.0                             | 0.355               | 0.560                            |
| Fair                        | 8                 | −0.58 (−1.25, 0.09)  | 0.088                | 89.5                            | ≤0.001              |                                  |
| Poor                        | 3                 | −0.64 (−2.40, 1.13)  | 0.479                | 99.3                            | ≤0.001              |                                  |



Table 2. continued

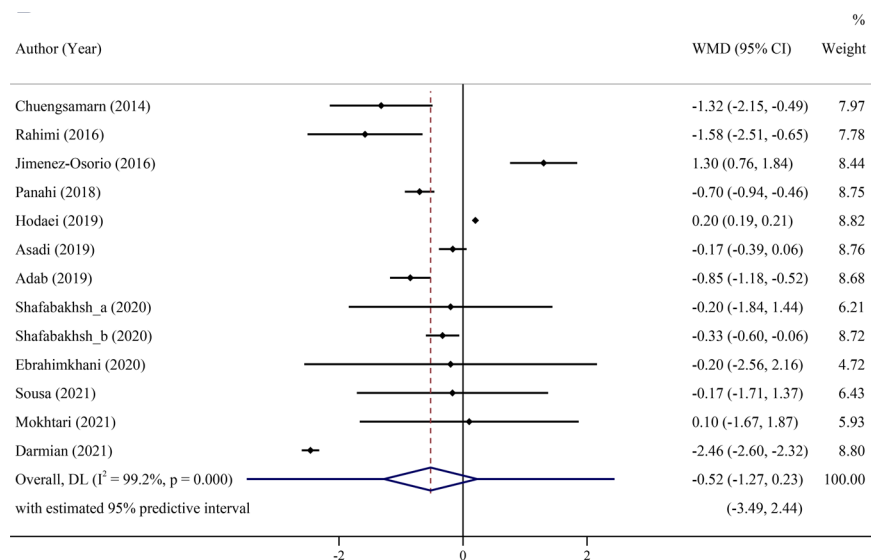
|                             | Number of studies | WMD (95% CI)         | P-value <sup>a</sup> | I <sup>2</sup> (%) <sup>b</sup> | P for heterogeneity | P between Subgroups <sup>c</sup> |
|-----------------------------|-------------------|----------------------|----------------------|---------------------------------|---------------------|----------------------------------|
| <b>Waist circumference</b>  |                   |                      |                      |                                 |                     |                                  |
| Overall                     | 5                 | −1.85 (−3.45, −0.24) | <b>0.024</b>         | 96.1                            | ≤0.001              |                                  |
| <b>Age</b>                  |                   |                      |                      |                                 |                     |                                  |
| <50 years                   | 1                 | −3.61 (−4.14, −3.08) | <b>≤0.001</b>        | 100.0                           | –                   | ≤0.001                           |
| ≥50 years                   | 4                 | −1.14 (−1.86, −0.43) | <b>0.002</b>         | 48.0                            | 0.123               |                                  |
| <b>Dose</b>                 |                   |                      |                      |                                 |                     |                                  |
| <1000 mg                    | 2                 | −1.31 (−2.18, −0.44) | <b>0.003</b>         | 0.0                             | 0.459               | 0.411                            |
| ≥1000 mg                    | 3                 | −2.33 (−4.62, −0.05) | <b>0.046</b>         | 98.0                            | ≤0.001              |                                  |
| <b>BMI</b>                  |                   |                      |                      |                                 |                     |                                  |
| <30 kg/m <sup>2</sup>       | 4                 | −1.95 (−4.00, 0.09)  | 0.061                | 97.0                            | ≤0.001              | 0.629                            |
| ≥30 kg/m <sup>2</sup>       | 1                 | −1.40 (−2.31, −0.49) | <b>0.002</b>         | –                               | –                   |                                  |
| <b>Duration</b>             |                   |                      |                      |                                 |                     |                                  |
| <12 weeks                   | 3                 | −1.93 (−3.91, 0.06)  | 0.057                | 98.0                            | ≤0.001              | 0.915                            |
| ≥12 weeks                   | 2                 | −1.76 (−4.17, 0.66)  | 0.155                | 45.4                            | 0.176               |                                  |
| <b>Type of Intervention</b> |                   |                      |                      |                                 |                     |                                  |
| Turmeric                    | 1                 | −3.61 (−4.14, −3.08) | <b>≤0.001</b>        | 100                             | –                   | ≤0.001                           |
| Unformulated curcumin       | 2                 | −1.50 (−3.34, 0.34)  | 0.111                | 74.4                            | 0.048               |                                  |
| High absorption curcumin    | 2                 | −1.31 (−2.18, −0.44) | <b>0.003</b>         | 0.0                             | 0.459               |                                  |
| <b>Location</b>             |                   |                      |                      |                                 |                     |                                  |
| Iran                        | 3                 | −1.93 (−3.91, 0.06)  | 0.057                | 98.0                            | ≤0.001              | 0.915                            |
| Other Countries             | 2                 | −1.76 (−4.17, 0.66)  | 0.155                | 45.4                            | 0.176               |                                  |
| <b>Risk of bias</b>         |                   |                      |                      |                                 |                     |                                  |
| Good                        | 1                 | −1.40 (−2.31, −0.49) | 0.002                | 0.0                             | –                   | ≤0.001                           |
| Fair                        | 3                 | −1.20 (−2.50, 0.10)  | 0.071                | 50.7                            | 0.131               |                                  |
| Poor                        | 1                 | −3.61 (−4.14, −3.08) | ≤0.001               | 100.0                           | –                   |                                  |

T2DM type 2 diabetes mellitus, WMD Weighted Mean Difference, CI confidence interval.

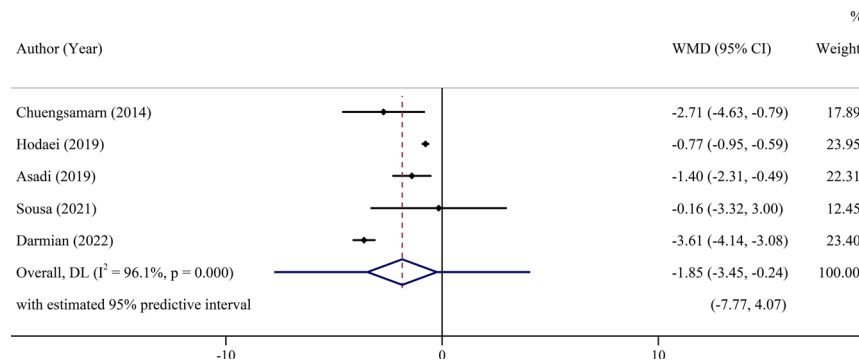
<sup>a</sup>P for heterogeneity, within subgroup.

<sup>b</sup>An I<sup>2</sup> value > 50% shows significant between-study heterogeneity.

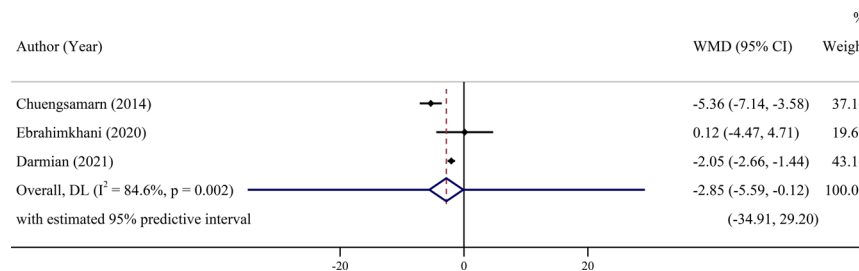
<sup>c</sup>P for heterogeneity, between subgroups.



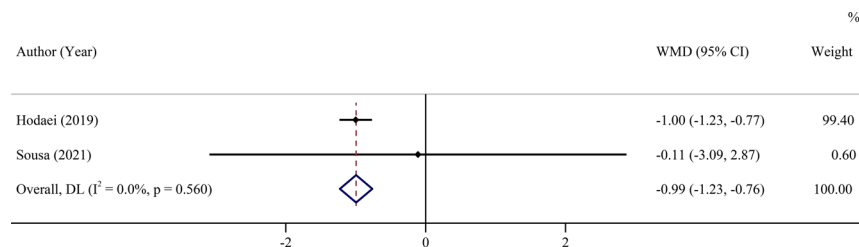
**Fig. 4 Forest plot of effect of turmeric/curcumin on body mass index in people with type 2 diabetes mellitus.** Horizontal lines represent 95% confidence intervals. Diamonds represent pooled estimates from random-effects analysis. Red dotted line represents null value. WMD weighted mean difference, DL DerSimonian & Laird method for random-effect meta-analysis.



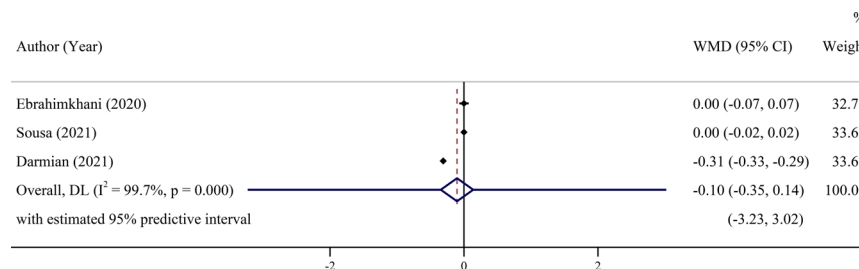
**Fig. 5 Forest plot of effect of turmeric/curcumin on waist circumference in people with type 2 diabetes mellitus.** Horizontal lines represent 95% confidence intervals. Diamonds represent pooled estimates from random-effects analysis. The red dotted line represents null value. WMD weighted mean difference, DL DerSimonian & Laird method for random-effect meta-analysis.



**Fig. 6 Forest plot of effect of turmeric/curcumin on fat mass% in people with type 2 diabetes mellitus.** Horizontal lines represent 95% confidence intervals. Diamonds represent pooled estimates from random-effects analysis. Red dotted line represents null value. WMD: weighted mean difference; DL: DerSimonian & Laird method for random-effect meta-analysis.



**Fig. 7 Forest plot of effect of turmeric/curcumin on hip circumference in people with type 2 diabetes mellitus.** Horizontal lines represent 95% confidence intervals. Diamonds represent pooled estimates from random-effects analysis. Red dotted line represents null value. WMD weighted mean difference, DL DerSimonian & Laird method for random-effect meta-analysis.

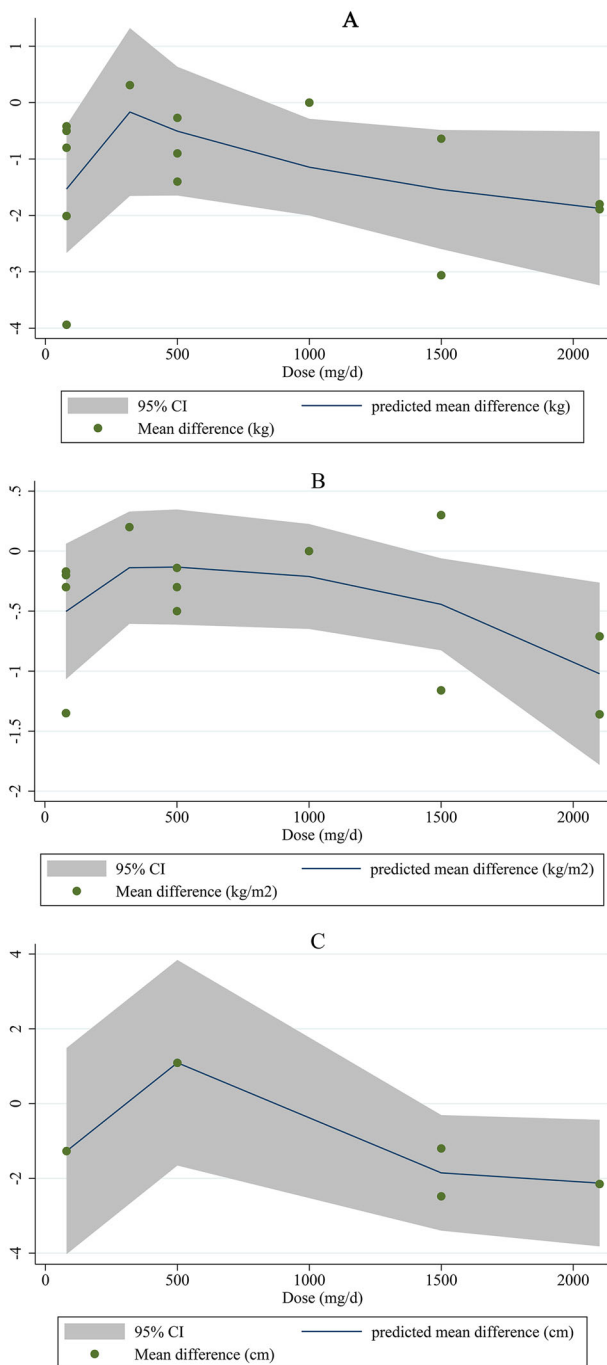


**Fig. 8 Forest plot of effect of turmeric/curcumin on waist-to-hip ratio in people with type 2 diabetes mellitus.** Horizontal lines represent 95% confidence intervals. Diamonds represent pooled estimates from random-effects analysis. The red dotted line represents null value. WMD weighted mean difference, DL DerSimonian & Laird method for random-effect meta-analysis.

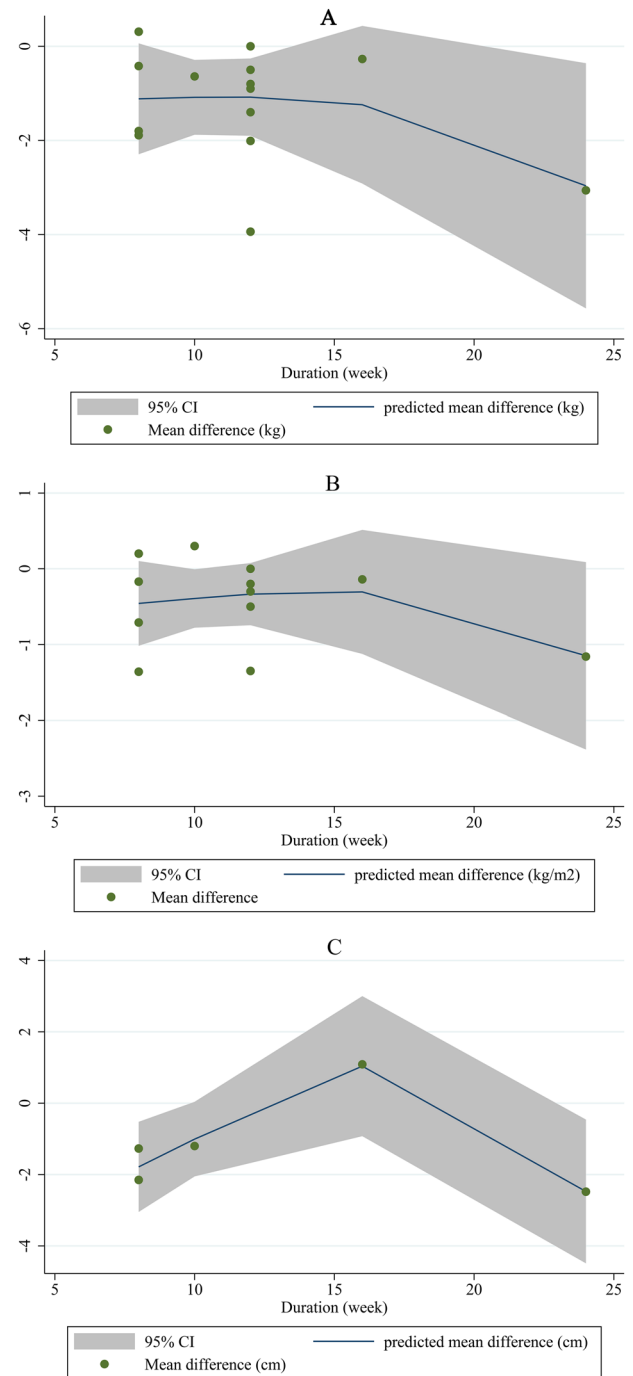
reported that supplementation with curcumin (500 mg/day) for 10 weeks decreased HC among individuals with T2DM [55].

This present systematic review and dose-response meta-analysis suggests supplementation with turmeric/curcumin does not have a significant impact on BMI in prediabetes and T2DM,

and is also not effective on WHR in T2DM. A previous RCT that lasted 120 days proves our result about WHR in T2DM [60]. Three systematic reviews and meta-analyses, in contrast with our study, reported curcumin or turmeric significantly lower BMI in adults and patients with metabolic-related diseases [22, 61]. However, it



**Fig. 9** Non-linear dose-response association between turmeric/curcumin dosage (mg/day) and mean difference in people with type 2 diabetes mellitus. **A** Body weight, **B** Body mass index, and **C** Waist circumference in type 2 diabetes mellitus population.



**Fig. 10** Non-linear dose-response association between turmeric/curcumin duration (week) and mean difference in people with type 2 diabetes mellitus. **A** body weight, **B** body mass index, **C** waist circumference in type 2 diabetes mellitus population.

should be noted that statistical significance was achieved in cases where supplementation extended for at least 12 weeks and when utilizing high-absorption forms of curcumin and turmeric.

As previously mentioned, the high-absorption form of curcumin and turmeric demonstrates a more pronounced impact on various outcomes. The limited bioavailability of curcumin has prompted the development of novel formulations [64]. When consumed curcumin orally, its bioavailability is influenced by metabolic processes. For example, adding glucuronide or sulfate groups to

curcumin in the liver are leading reason for its poor availability [65]. Curcuminoid in the form of turmeric exhibits greater bioavailability compared to pure curcumin, as turmeric is typically combined with other components such as fiber, turmeric oil, and starch [66]. Piperine, an alkaloid derived from black pepper, enhances curcumin bioavailability by inhibiting the glucuronidation process in the liver and intestine [67, 68]. The most effective methods for the synthesis of nano-curcumin involve ionic gelation and antisolvent precipitation, which enhance stability and

solubility. Moreover, incorporating curcumin into nanocarriers results in improved bioavailability, prolonged circulation, and retention in the body [69]. Phytosomal curcumin is produced by adding phospholipids to the ethanol solution of the hydroalcoholic extract of turmeric rhizomes, under reflux and stirring. The combination of curcumin with phosphatidylcholine enhances bioavailability and improves pharmacokinetics [64].

The findings of our meta-analysis indicated that three studies reported adverse events such as stomachache, itching, vertigo, constipation, hot flash, and nausea. This finding is consistent with a meta-analysis conducted on people at risk of cardiovascular disease [10].

The precise mechanism underlying the effects of curcumin on obesity has long been enigmatic. Curcumin plays a pivotal role in reducing angiogenesis and microvessel density within adipose tissue by downregulating the expression of Vascular Endothelial Growth Factor (VEGF) and VEGF Receptor 2 (VEGFR-2). Moreover, curcumin enhances fatty acid oxidation by elevating the expression of Carnitine Palmitoyltransferase-1 (CPT-1) [70]. Curcumin also activates AMP-activated protein kinase (AMPK), which impedes the differentiation of fat cells [71] and reduces adipocyte mass [72]. Additionally, curcumin induces the transcription of Uncoupling protein 1 (UCP1) through Peroxisome Proliferator-Activated Receptor- $\gamma$  (PPAR- $\gamma$ ) [73]. UCP1 is expressed in brown adipose tissue and has a primary role in thermogenesis and control of energy expenditure, which suggests these mechanisms have a major role in obesity pathogenesis [74]. Furthermore, this substance inhibits adipogenesis [75] and also prevents lipogenesis in the liver by suppressing the expression of carbohydrate response element binding protein (ChREBP) [76]. Curcumin exerts an inhibitory effect on 11 $\beta$ -HSD1, which this enzyme expressed in adipose tissue to activate glucocorticoid (GC). Increased GC interferes with the pathogenesis of central obesity [77].

The current systematic review and dose-response meta-analysis represent the first publication to assess the effects of turmeric/curcumin on anthropometric indices in individuals with prediabetes and T2DM. Our study had several strengths; First, we conducted a dose-response analysis to determine non-linear association between duration and dosage of supplementation with our outcomes. Second, we did not impose any language restrictions on the articles considered. Furthermore, our assessment of publication bias using Egger's test did not reveal any significant bias among the included RCTs. Additionally, we establish subgroup analyses to find the source of heterogeneity. Alongside these strengths, our study exhibits several limitations. The wide range of curcumin/turmeric dosages resulted in substantial heterogeneity. The heterogeneity observed between RCTs was influenced by differences in the form or dosage of curcumin/turmeric, various geographical regions, and variations in study duration. Finally, to address these limitations, future research should prioritize the conduct of high-quality RCTs with larger sample sizes and longer durations in diverse regions.

## CONCLUSIONS

In summary, our systematic review and dose-response meta-analysis revealed supplementation with turmeric/curcumin significantly diminished BW, WC, FM%, and HC in patients with T2DM. Furthermore, an advantageous effect of turmeric/curcumin consumption was observed in BW and WC among individuals with prediabetes.

## DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author, upon reasonable request.

## REFERENCES

- Reed J, Bain S, Kanamarlapudi V. A review of current trends with type 2 diabetes epidemiology, aetiology, pathogenesis, treatments and future perspectives. *Diab Metab Syndr Obes*. 2021;14:3567–602.
- Zheng Y, Ley SH, Hu FB. Global aetiology and epidemiology of type 2 diabetes mellitus and its complications. *Nat Rev Endocrinol*. 2018;14:88–98.
- Sun H, Saeedi P, Karuranga S, Pinkepank M, Ogurtsova K, Duncan BB, et al. IDF Diabetes Atlas: Global, regional and country-level diabetes prevalence estimates for 2021 and projections for 2045. *Diab Res Clin Pr*. 2022;183:109119.
- Khan T, Tzipas S, Wozniak G. Medical care expenditures for individuals with prediabetes: the potential cost savings in reducing the risk of developing diabetes. *Popul Health Manag*. 2017;20:389–96.
- Magliano DJ, Boyko EJ and IDF Diabetes Atlas 10th edition scientific committee, *IDF DIABETES ATLAS*. 2021, International Diabetes Federation, Brussels.
- Deshpande AD, Harris-Hayes M, Schootman M. Epidemiology of diabetes and diabetes-related complications. *Phys Ther*. 2008;88:1254–64.
- Devangan S, Varghese B, Johnny E, Gurram S, Adela R. The effect of *Gymnema sylvestre* supplementation on glycemic control in type 2 diabetes patients: A systematic review and meta-analysis. *Phytother Res*. 2021;35:6802–12.
- Unhapiatpong C, Polruang N, Shantavasinkul PC, Julanon N, Numthavaj P, Thakkestian A. The effect of curcumin supplementation on weight loss and anthropometric indices: an umbrella review and updated meta-analyses of randomized controlled trials. *Am J Clin Nutr*. 2023;117:1005–16.
- Kocaadam B, Şanlıer N. Curcumin, an active component of turmeric (*Curcuma longa*), and its effects on health. *Crit Rev Food Sci Nutr*. 2017;57:2889–95.
- Qin S, Huang L, Gong J, Shen S, Huang J, Ren H, et al. Efficacy and safety of turmeric and curcumin in lowering blood lipid levels in patients with cardiovascular risk factors: a meta-analysis of randomized controlled trials. *Nutr J*. 2017;16:68.
- Soleimani V, Sahebkar A, Hosseinzadeh H. Turmeric (*Curcuma longa*) and its major constituent (curcumin) as nontoxic and safe substances: Review. *Phytother Res*. 2018;32:985–95.
- Gao W, Chan JY, Wei W, Wong TS. Anti-cancer effects of curcumin on head and neck cancers. *Anticancer Agents Med Chem*. 2012;12:1110–6.
- Nelson KM, Dahlin JL, Bisson J, Graham J, Pauli GF, Walters MA. The essential medicinal chemistry of curcumin. *J Med Chem*. 2017;60:1620–37.
- Halegoua-DeMarzio D, Navarro V, Ahmad J, Avula B, Barnhart H, Barritt AS, et al. Liver injury associated with Turmeric-A growing problem: ten cases from the Drug-Induced Liver Injury Network [DILIN]. *Am J Med*. 2023;136:200–6.
- Daily JW, Yang M, Park S. Efficacy of turmeric extracts and curcumin for alleviating the symptoms of joint arthritis: a systematic review and meta-analysis of randomized clinical trials. *J Med Food*. 2016;19:717–29.
- Zeng L, Yang T, Yang K, Yu G, Li J, Xiang W, et al. Curcumin and *Curcuma longa* extract in the treatment of 10 types of autoimmune diseases: a systematic review and meta-analysis of 31 randomized controlled trials. *Front Immunol*. 2022;13:896476.
- Ferguson JJA, Abbott KA, Garg ML. Anti-inflammatory effects of oral supplementation with curcumin: a systematic review and meta-analysis of randomized controlled trials. *Nutr Rev*. 2021;79:1043–66.
- Fusar-Poli L, Vozza L, Gabbiadini A, Vanella A, Concas I, Tinacci S, et al. Curcumin for depression: a meta-analysis. *Crit Rev Food Sci Nutr*. 2020;60:2643–53.
- Dehzad MJ, Ghalandari H, Amini MR, Askarpour M. Effects of curcumin/turmeric supplementation on liver function in adults: A GRADE-assessed systematic review and dose-response meta-analysis of randomized controlled trials. *Complement Ther Med*. 2023;74:102952.
- Pricci M, Girardi B, Giorgio F, Losurdo G, Ierardi E, Di Leo A. Curcumin and colorectal cancer: from basic to clinical evidences. *Int J Mol Sci*. 2020;21:2364.
- Jabczyk M, Nowak J, Hudzik B, Zubelewicz-Szkodzińska B. Curcumin in metabolic health and disease. *Nutrients*. 2021;13:4440.
- Sun Z, Wei X, Bai J, Li W, Yang J, Deng Z et al., The effects of curcumin on anthropometric and cardiometabolic parameters of patients with metabolic related diseases: a systematic review and dose-effect meta-analysis of randomized controlled trials. *Cri Rev Food Sci Nutr*, 2022: p. 1-17.
- Seo KI, Choi MS, Jung UJ, Kim HJ, Yeo J, Jeon SM, et al. Effect of curcumin supplementation on blood glucose, plasma insulin, and glucose homeostasis related enzyme activities in diabetic db/db mice. *Mol Nutr Food Res*. 2008;52:995–1004.
- Kasprzak-Drozd K, Oniszczuk T, Gancarz M, Kondracka A, Rusinek R, Oniszczuk A. Curcumin and weight loss: does it work? *Int J Mol Sci*. 2022;23:639.
- Akbari M, Lankarani KB, Tabrizi R, Ghayour-Mobarhan M, Peymani P, Ferns G, et al. The effects of curcumin on weight loss among patients with metabolic syndrome and related disorders: a systematic review and meta-analysis of randomized controlled trials. *Front Pharm*. 2019;10:649.
- Mousavi SM, Milajerdi A, Varkaneh HK, Gorjipour MM, Esmaillzadeh A. The effects of curcumin supplementation on body weight, body mass index and waist

- circumference: a systematic review and dose-response meta-analysis of randomized controlled trials. *Crit Rev Food Sci Nutr*. 2020;60:171–80.
27. Ashtary-Larky D, Rezaei Kelishadi M, Bagheri R, Moosavian SP, Wong A, Davoodi SH, et al. The effects of nano-curcumin supplementation on risk factors for cardiovascular disease: A GRADE-assessed systematic review and meta-analysis of clinical trials. *Antioxidants*. 2021;10:1015.
  28. Altobelli E, Angeletti PM, Marziliano C, Mastrodomenico M, Giuliani AR, Petrocelli R. Potential therapeutic effects of curcumin on glycemic and lipid profile in uncomplicated type 2 diabetes—a meta-analysis of randomized controlled trial. *Nutrients*. 2021;13:404. <https://doi.org/10.3390/nu13020404>
  29. Zhang T, He Q, Liu Y, Chen Z, Hu H. Efficacy and safety of curcumin supplement on improvement of insulin resistance in people with type 2 diabetes mellitus: a systematic review and meta-analysis of randomized controlled trials. *Evid-Based Complement Altern Med*. 2021;2021:4471944.
  30. Tian J, Feng B, Tian Z. The effect of curcumin on lipid profile and glycemic status of patients with type 2 diabetes mellitus: a systematic review and meta-analysis. *Evid-Based Complement Altern Med*. 2022;2022:8278744.
  31. Chuengsamarn S, Rattanamongkolgul S, Luechapudiporn R, Phisalaphong C, Jirawatnotai S. Curcumin extract for prevention of type 2 diabetes. *Diab Care*. 2012;35:2121–7.
  32. Funamoto M, Shimizu K, Sunagawa Y, Katanasaka Y, Miyazaki Y, Kakeya H, et al. Effects of highly absorbable curcumin in patients with impaired glucose tolerance and non-insulin-dependent diabetes mellitus. *J Diab Res*. 2019;2019:8208237.
  33. Thota RN, Rosato JL, Dias CB, Burrows TL, Martins RN, Garg ML. Dietary supplementation with curcumin reduce circulating levels of glycogen Synthase Kinase-3 $\beta$  and Islet Amyloid Polypeptide in adults with high risk of type 2 diabetes and Alzheimer's disease. *Nutrients*. 2020;12:1032.
  34. Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ*. 2021;372:n71.
  35. Higgins JPT, Altman DG, Gøtzsche PC, Jüni P, Moher D, Oxman AD, et al. The Cochrane Collaboration's tool for assessing risk of bias in randomised trials. *BMJ*. 2011;343:d5928.
  36. Borenstein M, Hedges LV, Higgins JP, Rothstein HR *Introduction to meta-analysis*. John Wiley & Sons. (2021).
  37. Hozo SP, Djulbegovic B, Hozo I. Estimating the mean and variance from the median, range, and the size of a sample. *BMC Med Res Methodol*. 2005;5:13.
  38. DerSimonian R, Laird N. Meta-analysis in clinical trials. *Control Clin Trials*. 1986;7:177–88.
  39. Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ et al. *Cochrane handbook for systematic reviews of interventions*. John Wiley & Sons. (2019).
  40. Egger M, Davey Smith G, Schneider M, Minder C. Bias in meta-analysis detected by a simple, graphical test. *BMJ*. 1997;315:629–34.
  41. Begg CB, Mazumdar M. Operating characteristics of a rank correlation test for publication bias. *Biometrics*. 1994;50:1088–101.
  42. Tobias A. Assessing the influence of a single study in the meta-analysis estimate. *Stata Tech Bulletin*. 1999. 8.
  43. Guyatt GH, Oxman AD, Kunz R, Falck-Ytter Y, Vist GE, Liberati A, et al. Going from evidence to recommendations. *BMJ*. 2008;336:1049–51.
  44. Rahimi HR, Mohammadpour AH, Dastani M, Jaafari MR, Abnous K, Ghayour Mobarhan M, et al. The effect of nano-curcumin on HbA1c, fasting blood glucose, and lipid profile in diabetic subjects: a randomized clinical trial. *Avicenna J Phytomed*. 2016;6:567–77.
  45. Asadi S, Gholami MS, Siassi F, Qorbani M, Khamoshian K, Sotoudeh G. Nano curcumin supplementation reduced the severity of diabetic sensorimotor polyneuropathy in patients with type 2 diabetes mellitus: A randomized double-blind placebo-controlled clinical trial. *Complement Ther Med*. 2019;43:253–60.
  46. Shafabakhsh R, Asemi Z, Reiner Z, Soleimani A, Aghadavod E, Bahmani F. The effects of Nano-curcumin on metabolic status in patients with diabetes on hemodialysis, a randomized, double blind, placebo-controlled trial. *Iran J Kidney Dis*. 2020;14:290–9.
  47. Mokhtari M, Razzaghi R, Momen-Heravi M. The effects of curcumin intake on wound healing and metabolic status in patients with diabetic foot ulcer: A randomized, double-blind, placebo-controlled trial. *Phytother Res*. 2021;35:2099–107.
  48. Asghari KM, Saleh P, Salekzamani Y, Dolatkah N, Aghamohammadzadeh N, Hashemian M. The effect of curcumin and high-content eicosapentaenoic acid supplementations in type 2 diabetes mellitus patients: a double-blinded randomized clinical trial. *Nutr Diab*. 2024;14:14.
  49. Adab Z, Eghtesadi S, Vafa MR, Heydari I, Shojaii A, Haqqani H, et al. Effect of turmeric on glycemic status, lipid profile, hs-CRP, and total antioxidant capacity in hyperlipidemic type 2 diabetes mellitus patients. *Phytother Res*. 2019;33:1173–81.
  50. Darmian M, Hoseini R, Amiri E, Golshani S. How combined and separate aerobic training and turmeric supplementation alter lipid profile and glycemic status? A clinical trial in middle-aged females with Type 2 Diabetes and Hyperlipidemia. *Int Cardiovasc Res J*. 2021;5:111–8.
  51. Darmian MA, Hoseini R, Amiri E, Golshani S. Downregulated hs-CRP and MAD, upregulated GSH and TAC, and improved metabolic status following combined exercise and turmeric supplementation: a clinical trial in middle-aged women with hyperlipidemic type 2 diabetes. *J Diab Metab Disord*. 2022;21:275–83.
  52. Jiménez-Osorio AS, García-Niño WR, González-Reyes S, Álvarez-Mejía AE, Guerra-León S, Salazar-Segovia J, et al. The effect of dietary supplementation with Curcumin on Redox Status and Nrf2 activation in patients with nondiabetic or diabetic proteinuric chronic kidney disease: a pilot study. *J Ren Nutr*. 2016;26:237–44.
  53. Cicero AFG, Sahebkar A, Fogacci F, Bove M, Giovannini M, Borghi C. Effects of phytosomal curcumin on anthropometric parameters, insulin resistance, cortisolemia and non-alcoholic fatty liver disease indices: a double-blind, placebo-controlled clinical trial. *Eur J Nutr*. 2020;59:477–83.
  54. Chuengsamarn S, Rattanamongkolgul S, Phonrat B, Tungtrongchitr R, Jirawatnotai S. Reduction of atherogenic risk in patients with type 2 diabetes by curcuminoid extract: a randomized controlled trial. *J Nutr Biochem*. 2014;25:144–50.
  55. Hodaie H, Adibian M, Nikpayam O, Hedayati M, Sohrab G. The effect of curcumin supplementation on anthropometric indices, insulin resistance and oxidative stress in patients with type 2 diabetes: a randomized, double-blind clinical trial. *Diabetol Metab Syndr*. 2019;11:41.
  56. Shafabakhsh R, Mobini M, Raygan F, Aghadavod E, Ostadmohammadi V, Amirani E, et al. Curcumin administration and the effects on psychological status and markers of inflammation and oxidative damage in patients with type 2 diabetes and coronary heart disease. *Clin Nutr ESPEN*. 2020;40:77–82.
  57. Ebrahimkhani S, Ghavamzadeh S, Mehdizadeh A. The effects of vitamin D and curcuminoids supplementation on anthropometric measurements and blood pressure in type 2 diabetic patients with coexisting hypovitaminosis D: A double-blind, placebo-controlled randomized clinical trial. *Clin Nutr ESPEN*. 2020;37:178–86.
  58. Karandish M, Mozaffari-Khosravi H, Mohammadi SM, Cheraghian B, Azhdari M. Curcumin and zinc co-supplementation along with a loss-weight diet can improve lipid profiles in subjects with prediabetes: a multi-arm, parallel-group, randomized, double-blind placebo-controlled phase 2 clinical trial. *Diabetol Metab Syndr*. 2022;14:22.
  59. Panahi Y, Khalili N, Sahebi E, Namazi S, Simental-Mendia LE, Majeed M, et al. Effects of Curcuminoids plus piperine on glycemic, hepatic and inflammatory biomarkers in patients with type 2 diabetes mellitus: a randomized double-blind placebo-controlled trial. *Drug Res (Stuttg)*. 2018;68:403–9.
  60. Sousa DF, Araújo MFM, de Mello VD, Damasceno MMC, Freitas R. Cost-effectiveness of passion Fruit Albedo versus turmeric in the glycemic and lipaemic control of people with type 2 diabetes: randomized clinical trial. *J Am Coll Nutr*. 2021;40:679–88.
  61. Dehzad MJ, Ghalandari H, Nouri M, Askarpour M. Effects of curcumin/turmeric supplementation on obesity indices and adipokines in adults: A grade-assessed systematic review and dose-response meta-analysis of randomized controlled trials. *Phytother Res*. 2023;37:1703–28.
  62. Jafarirad S, Mansoori A, Adineh A, Panahi Y, Hadi A, Goodarzi R. Does Turmeric/curcumin supplementation change anthropometric indices in patients with non-alcoholic fatty liver disease? A systematic review and meta-analysis of randomized controlled trials. *Clin Nutr Res*. 2019;8:196–208.
  63. Feller S, Boeing H, Pischon T. Body mass index, waist circumference, and the risk of type 2 diabetes mellitus: implications for routine clinical practice. *Dtsch Arztebl Int*. 2010;107:470–6.
  64. Mirzaei H, Shakeri A, Rashidi B, Jalili A, Banikazemi Z, Sahebkar A. Phytosomal curcumin: A review of pharmacokinetic, experimental and clinical studies. *Biomed Pharmacother*. 2017;85:102–12.
  65. Sabet S, Rashidinejad A, Melton LD, McGillivray DJ. Recent advances to improve curcumin oral bioavailability. *Trends Food Sci Technol*. 2021;110:253–66.
  66. Ahmed Nasef N, Loveday SM, Golding M, Martins RN, Shah TM, Clarke M, et al. Food matrix and co-presence of turmeric compounds influence bioavailability of curcumin in healthy humans. *Food Funct*. 2019;10:4584–92.
  67. Dei Cas M, Ghidoni R. Dietary curcumin: correlation between bioavailability and health potential. *Nutrients*. 2019;11:2147.
  68. Heidari H, Bagherniya M, Majeed M, Sathiyapalan T, Jamialahmadi T, Sahebkar A. Curcumin-piperine co-supplementation and human health: A comprehensive review of preclinical and clinical studies. *Phytother Res*. 2023;37:1462–87.
  69. Karthikeyan A, Senthil N, Min T. Nanocurcumin: A promising candidate for therapeutic applications. *Front Pharmacol*. 2020;11:487.
  70. Ejaz A, Wu D, Kwan P, Meydani M. Curcumin inhibits adipogenesis in 3T3-L1 adipocytes and angiogenesis and obesity in C57/BL mice. *J Nutr*. 2009;139:919–25.
  71. Lee YK, Lee WS, Hwang JT, Kwon DY, Surh YJ, Park OJ. Curcumin exerts anti-differentiation effect through AMPK $\alpha$ -PPAR- $\gamma$  in 3T3-L1 adipocytes and



- antiproliferatory effect through AMPK $\alpha$ -COX-2 in cancer cells. *J Agric Food Chem.* 2009;57:305–10.
72. Islam T, Koboziev I, Albracht-Schulte K, Mistretta B, Scoggins S, Yosofvand M, et al. Curcumin reduces adipose tissue inflammation and alters gut microbiota in diet-induced obese male mice. *Mol Nutr Food Res.* 2021;65:e2100274.
  73. Song Z, Revelo X, Shao W, Tian L, Zeng K, Lei H, et al. Dietary curcumin intervention targets mouse white adipose tissue inflammation and brown adipose tissue UCP1 Expression. *Obesity.* 2018;26:547–58.
  74. Brondani LA, Assmann TS, Duarte GC, Gross JL, Canani LH, Crispim D. The role of the uncoupling protein 1 (UCP1) on the development of obesity and type 2 diabetes mellitus. *Arq Bras Endocrinol Metab.* 2012;56:215–25.
  75. Wu LY, Chen CW, Chen LK, Chou HY, Chang CL, Juan CC. Curcumin Attenuates adipogenesis by inducing preadipocyte apoptosis and inhibiting adipocyte differentiation. *Nutrients.* 2019;11:2307.
  76. Hao M, Chu Y, Lei J, Yao Z, Wang P, Chen Z, et al. Pharmacological mechanisms and clinical applications of curcumin: update. *Aging Dis.* 2023;14:716–49.
  77. Hu GX, Lin H, Lian QQ, Zhou SH, Guo J, Zhou HY, et al. Curcumin as a potent and selective inhibitor of 11 $\beta$ -hydroxysteroid dehydrogenase 1: improving lipid profiles in high-fat-diet-treated rats. *PLoS One.* 2013;8:e49976.

## AUTHOR CONTRIBUTIONS

MMB, PA, and LA designed the study. MMB and AS performed a literature search and data extraction. Data analysis was done by MMB, PA and MM. MMB and AS wrote the paper. PA, MM, ENE, and LA revised the final version of the paper. All authors read and approved the final manuscript.

## FUNDING

This study is supported by Tehran University of Medical Sciences (Grant number: IR.TUMS.EMRI.REC.1401.145).

## COMPETING INTERESTS

The authors declare no competing interests.

## ETHICS APPROVAL AND CONSENT TO PARTICIPATE

As the present study is a secondary analysis of data from previously published research, it did not require ethical approval.

## ADDITIONAL INFORMATION

**Supplementary information** The online version contains supplementary material available at <https://doi.org/10.1038/s41387-025-00386-7>.

**Correspondence** and requests for materials should be addressed to Leila Azadbakht.

**Reprints and permission information** is available at <http://www.nature.com/reprints>

**Publisher's note** Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.



**Open Access** This article is licensed under a Creative Commons

Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

© The Author(s) 2025